MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

PROJECTILE, 8/55 AP, MK 21 AND MODS

HC, MK 24 OR MK 25 AND MODS

FLEET ISSUE UNIT LOAD

MIL-STD-1320-44

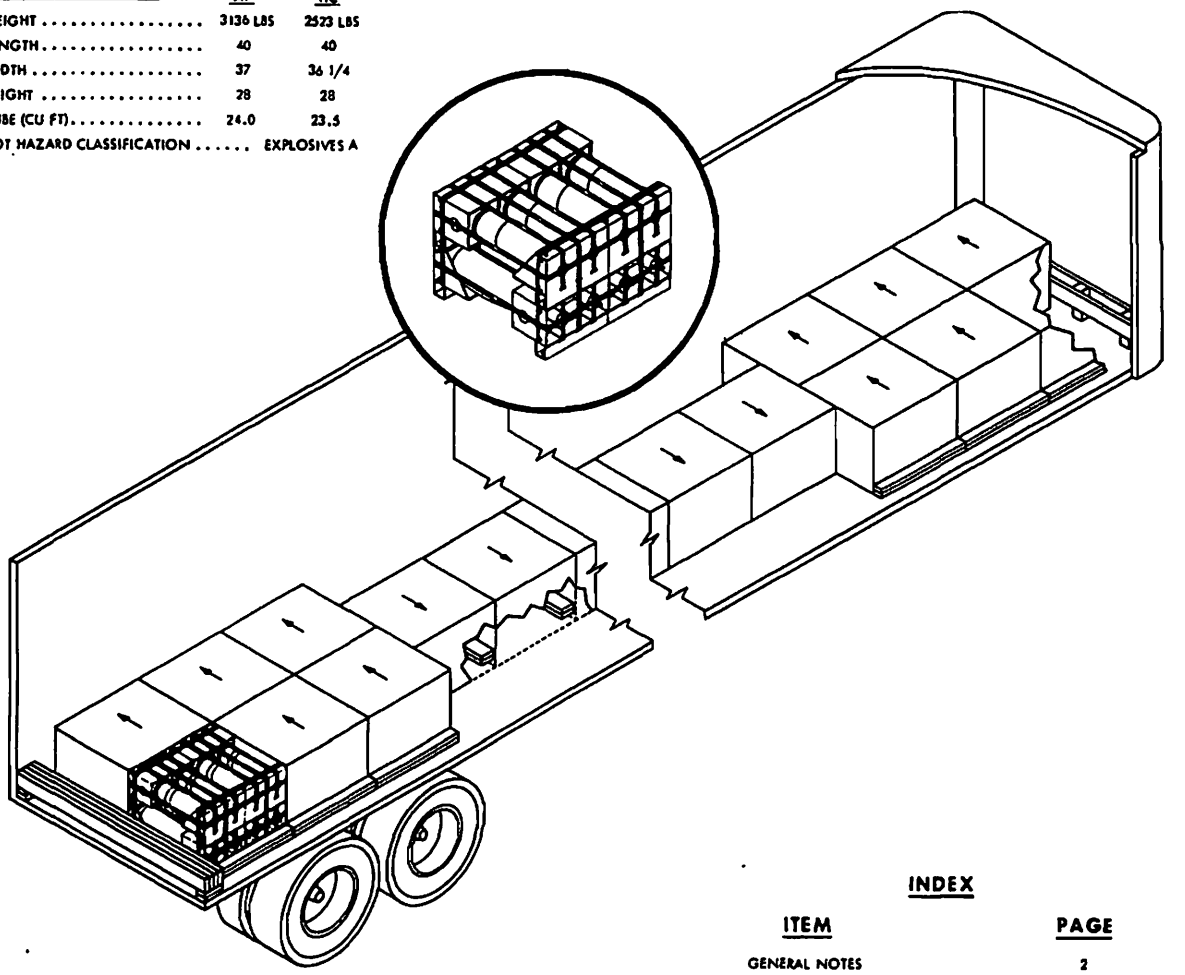
(NAVY)

2 APRIL 1975

SUPERSEDING
WR-51/44
9 OCTOBER 1967

UNIT LOAD DATA

	AP	HC
WEIGHT	3136 LBS	2523 LBS
LENGTH	40	40
WIDTH	37	36 1/4
HEIGHT	28	28
CUBE (CU FT)	24.0	23.5
DOT HAZARD CLASSIFICATION	EXPLOSIVES A	



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236.

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray, Mitre 3/26/75
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ORIGINATOR T. C. Wingi
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NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF 8"/55 MK 21 AP PROJECTILES AND MK 24 OR MK 25 HC PROJECTILES WHEN PALLETIZED IN ACCORDANCE WITH WR-54/123 OR WR-54/124.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH WOOD FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS. THE TRAILERS SHALL HAVE THE TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. TO FIND THE CORRECT LOAD PATTERN TO USE:
   a. DETERMINE THE TYPE OR MK OF PROJECTILE TO BE LOADED (AP, MK 21 OR HC, MK 24 OR MK 25).
   b. SELECT THE CORRECT TABLE FOR THE TYPE OR MK OF PROJECTILE BEING SHIPPED. (TABLE I OR TABLE II).
   c. DETERMINE THE LENGTH OF THE TRAILER BEING LOADED. (WHEN USING 45 FT TRAILERS, SEE FTL NOTE 2.)
   d. FIND THE LOAD PATTERN NUMBER IN THE SELECTED TABLE USING THE NUMBER OF UNIT LOADS BEING SHIPPED AND THE LENGTH OF THE TRAILER BEING LOADED.
   e. USING THE LOAD PATTERN NUMBER, ENTER TABLE III AND FIND THE CORRECT PATTERN TO BE USED.
   f. LOAD, BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 4 & 5.
4. THE CORRECT REAR BLOCKING TO BE USED DEPENDS UPON THE DISTANCE BETWEEN THE REAR OF THE LADING AND THE TRAILER DOORS, WHEN CLOSED. WHEN THAT DISTANCE IS:

   LESS THAN 12 INCHES

   INSTALL REAR BLOCKING ASSEMBLY DETAIL A, PAGE 2, AS SHOWN ON PAGE 4.

   CAUTION

   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

   12 TO 36 INCHES

   INSTALL REAR BLOCKING ASSEMBLY, DETAIL F, PAGE 5, IN LIEU OF REAR BLOCKING SHOWN ON PAGE 4.

   CAUTION

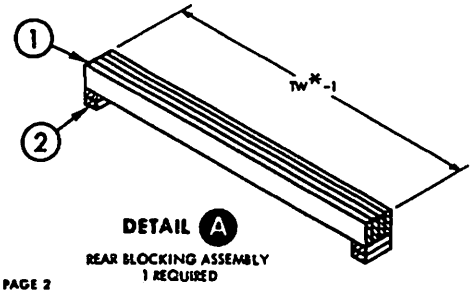
   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

   GREATER THAN 36 INCHES

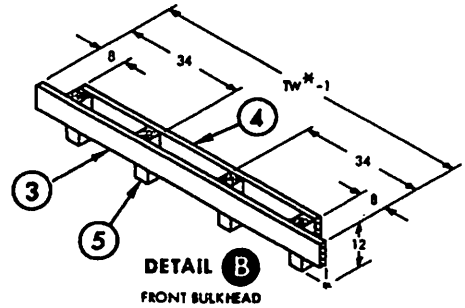
   INSTALL REAR BLOCKING AS SHOWN FOR LTL ON PAGE 5.
5. CLEAT, PIECE 6, IS USED WHEN TRAILER HAS METAL FLOORS WITH WOOD NAILING STRIPS. USE ONE CLEAT (DOUBLED) AT EACH END OF THE SINGLE ROW AND ONE CLEAT (DOUBLED) WHERE TWO UNIT LOADS BUTT TOGETHER. NAIL CLEAT TO NAILING STRIP. DO NOT NAIL INTO METAL FLOOR.
6. WHEN THE TRAILER HAS A NAILABLE FLOOR, AN ALTERNATE FOR CLEAT, PIECE 6, IS 2 x 4 DOUBLED SLEEPER ON EACH SIDE OF THE SINGLE ROW INSTALLED IN THE SAME MANNER AS SLEEPER, PIECE 7.
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVE A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT.
10. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40, 42, 44 & 45 FT TRAILERS (VAN)

1. A FULL TRUCKLOAD CONSISTS OF:
   a. 17 UNIT LOADS OF MK 24 OR MK 25 HC PROJECTILES OR 13 UNIT LOADS OF MK 21 AP PROJECTILES WHEN LOADED INTO A STANDARD WEIGHT TRACTOR-TRAILER (TARE VEHICLE WEIGHT 27,500 LBS MAXIMUM).
   b. 18 UNIT LOADS OF MK 24 OR MK 25HC PROJECTILES OR 14 UNIT LOADS OF MK 21 AP PROJECTILES WHEN LOADED INTO A LIGHTWEIGHT TRACTOR-TRAILER (TARE VEHICLE WEIGHT, 25,500 LBS MAXIMUM).
2. WHEN LOADING 45 FT TRAILERS, USE THE SAME LOAD PLAN AND PROCEDURES AS SPECIFIED FOR THE 44 FT TRAILERS, INCREASING THE THICKNESS OF THE FRONT BULKHEAD BY SIX INCHES.



DETAIL A
REAR BLOCKING ASSEMBLY
1 REQUIRED



DETAIL B
FRONT BULKHEAD
1 REQUIRED

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TABLE I

HC PROJECTILE MK 24 OR MK 25 AND MODS

NO. OF UNIT LOADS TO BE SHIPPED	FTL OR LTL	LOAD PATTERN NO. TRAILER LENGTH			TRUCKLOAD WEIGHT (LBS)
		40	42	44	
18	FTL	3*	2*	1*	45,414
17	FTL	6	5	4	42,891
16	LTL	9	8	7	40,368
15	LTL	11	10	10	37,845
14	LTL	13	12	12	35,332
13	LTL	15	14	14	32,799
12	LTL	17	16	16	30,276
11	LTL	18	18	18	27,753
10 OR LESS	USE SINGLE ROW PATTERN				

TABLE II

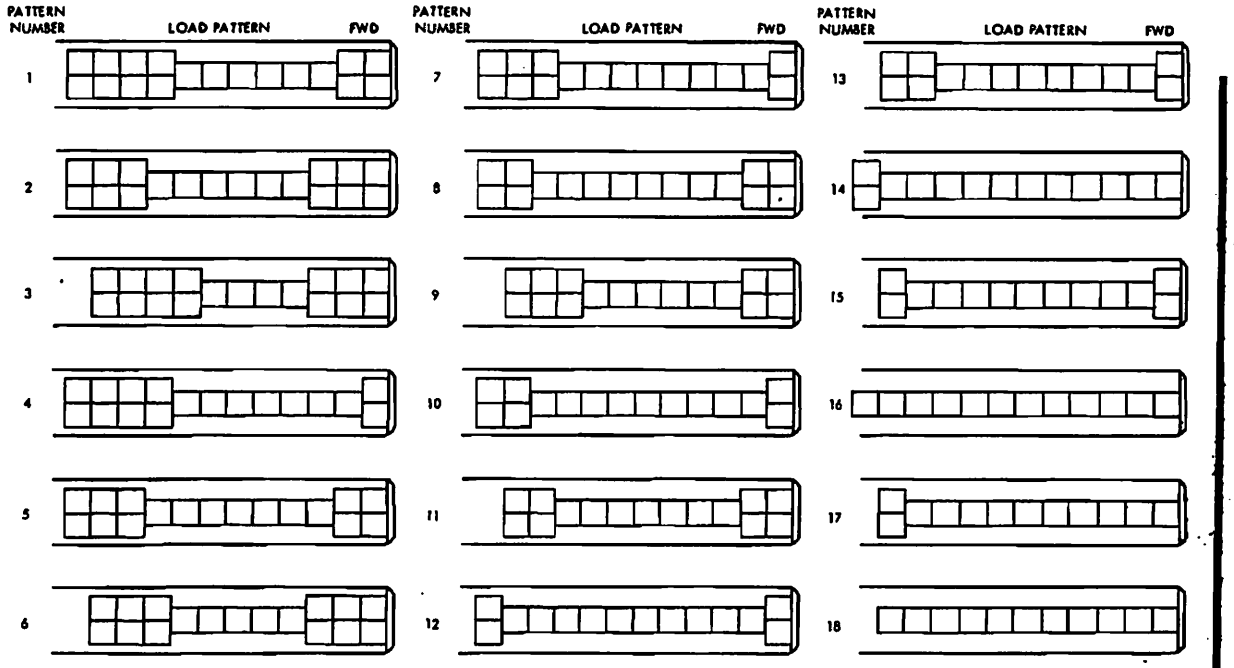
AP PROJECTILE, MK 21 AND MODS

NO. OF UNIT LOADS TO BE SHIPPED	FTL OR LTL	LOAD PATTERN NO. TRAILER LENGTH			TRUCKLOAD WEIGHT (LBS)
		40	42	44	
14	FTL	13*	12*	12*	43,904
13	FTL	15	14	14	40,768
12	LTL	17	16	16	37,632
11	LTL	18	18	18	34,496
10 OR LESS	USE SINGLE ROW PATTERN				

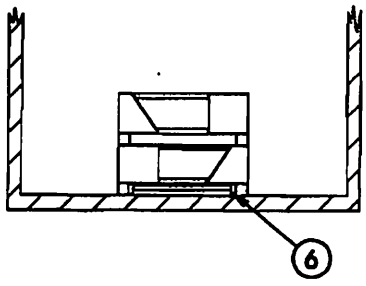
WARNING

* THIS LOAD IS SUITABLE FOR LIGHTWEIGHT TRACTOR-TRAILERS ONLY. IF LOADED ON A STANDARD WEIGHT TRACTOR-TRAILER, THE AXLE WEIGHTS AND THE GROSS VEHICLE WEIGHT WILL BE IN EXCESS OF THE LIMITS OF 32,000 LBS FOR TANDEM AXLES AND 73,280 LBS GROSS VEHICLE WEIGHT. SEE FTL NOTE 1.

TABLE III



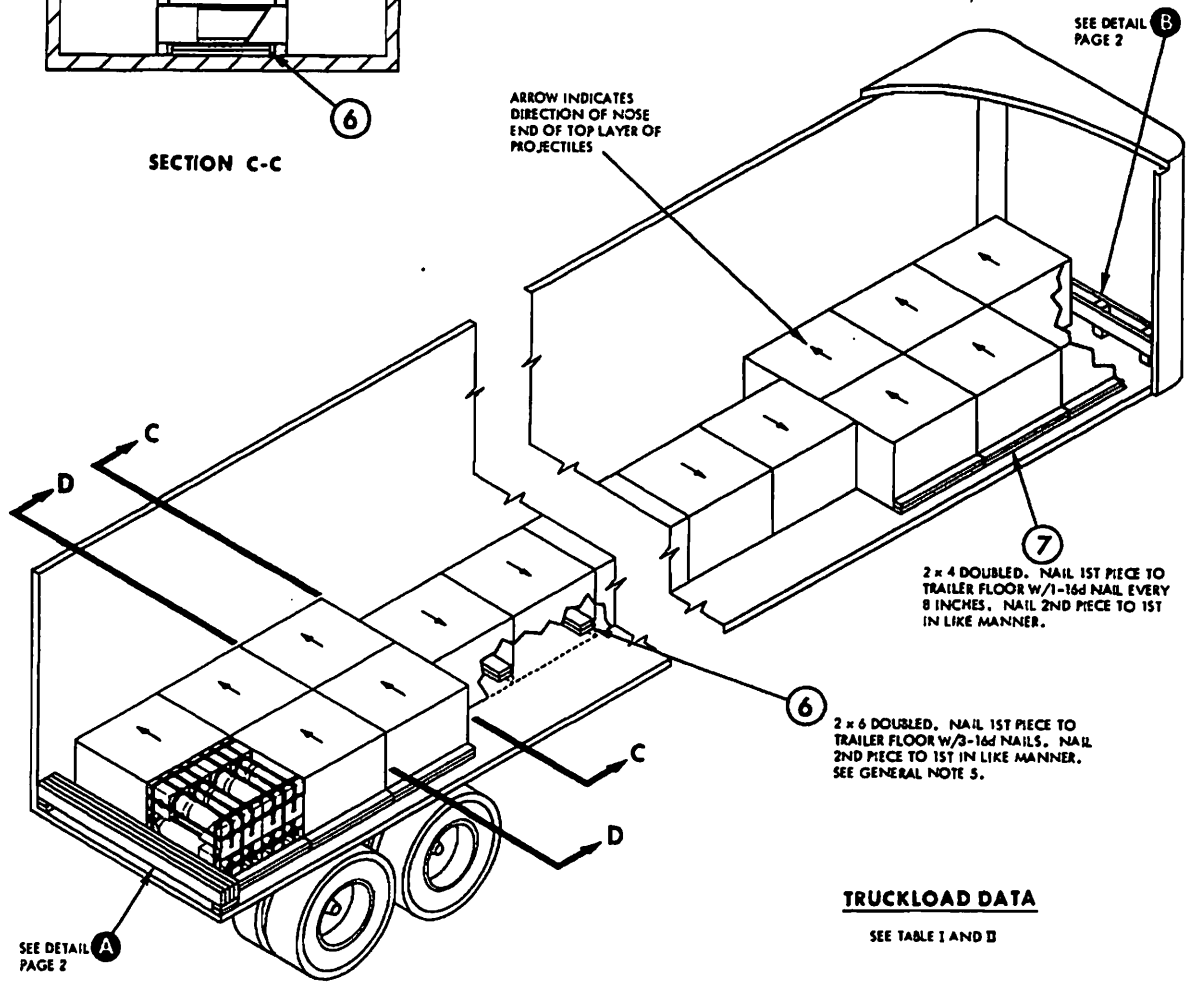
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SECTION C-C

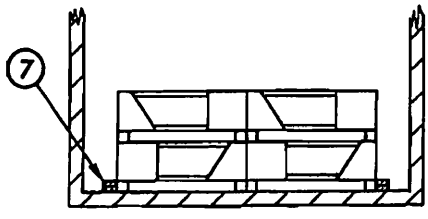
NOTE

THIS DRAWING ILLUSTRATES A TRAILER WITH METAL FLOORS WITH WOOD NAILING STRIPS. SEE GENERAL NOTES 4 AND 5.



TRUCKLOAD DATA

SEE TABLE I AND II



SECTION D-D

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS AND NAILING DATA				NAILS	
7	SLEEPER	2 x 4 x 40	AS REQD	SEE FIELD NOTE		
6	CLEAT	2 x 6 x UNIT LOAD WIDTH - 8	AS REQD	SEE FIELD NOTE		
5	UPRIGHT	4 x 4 x 12	4	SEE 3 & 4	-	-
4	FWD CROSSMEMBER	2 x 6 TO SUIT	1	5	3/JOINT	12d
3	AFT CROSSMEMBER	2 x 6 x TW*-1	1	5	3/JOINT	12d
2	SUPPORT	2 x 4 TO SUIT	4	1 2	2/JOINT 2	12d 12d
1	CROSSMEMBER	THICKNESS TO SUIT x 6 x TW*-1	AS REQD	LAMINATE	6/JOINT	AS REQD

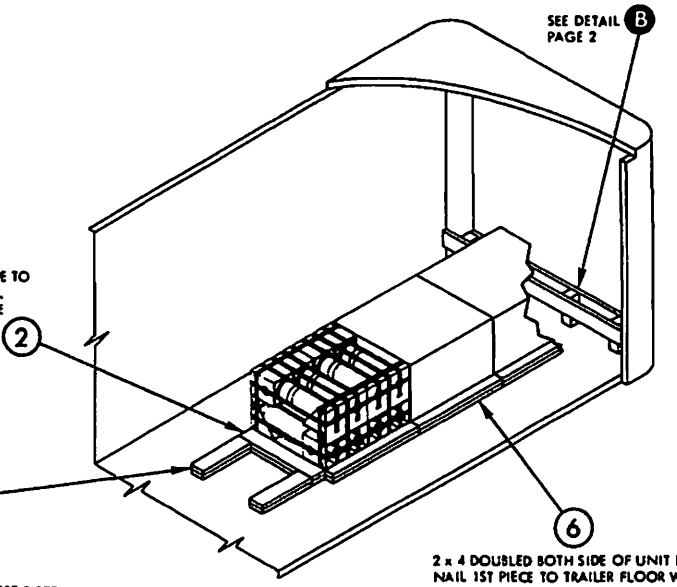
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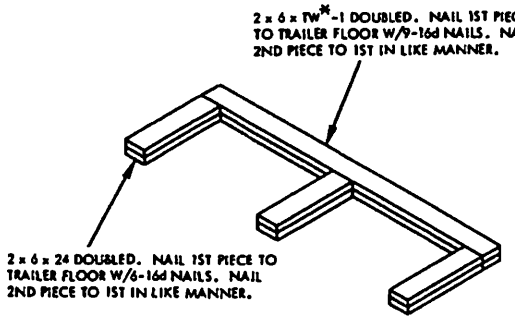
LTL ALL TRAILERS

1. WHEN SHIPPING MORE THAN 10 UNIT LOADS, USE LOAD PATTERNS PRESCRIBED BY TABLES I AND II. BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGE 4.
2. WHEN SHIPPING 10 OR LESS UNIT LOADS, USE A SINGLE ROW PATTERN. BLOCK AND BRACE USING THE PRINCIPLES SHOWN BELOW.
3. WHEN A TRAILER HAS METAL FLOORS WITH NAILING STRIPS, SUBSTITUTE DETAIL E FOR REAR BLOCKING AND SUBSTITUTE CLEATS, PIECE 6 PAGE 4 FOR SLEEPERS.

NOTE

THIS DRAWING ILLUSTRATES TRAILERS WITH WOOD FLOORS. SEE NOTE 3 ABOVE.





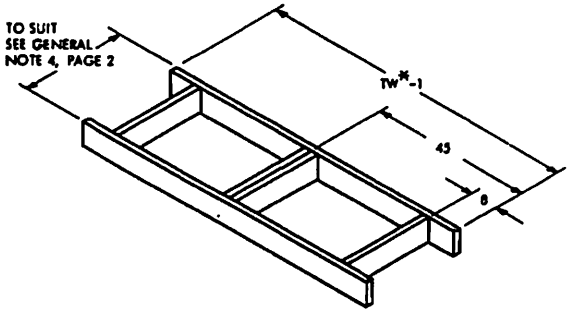
DETAIL E

REAR BLOCKING
(FOR METAL FLOORS WITH NAILING STRIPS)

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
6	SLEEPER	2 x 4 x 40	12	SEE FIELD NOTE		
5	VERTICAL	4 x 4 x 12	4	SEE 3 & 4	-	-
4	AFT CROSSMEMBER	2 x 6 TO SUIT	1	5	3/JOINT	12d
3	FWD CROSSMEMBER	2 x 6 x TW*-1	1	5	3/JOINT	12d
2	CROSSMEMBER	2 x 6 TO SUIT	2	SEE FIELD NOTE		
1	BACKUP CLEAT	2 x 6 x 30	4	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA



DETAIL F

REAR BLOCKING ASSEMBLY
(SPACE AT REAR 12 TO 36 INCHES)

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8140-N289)

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